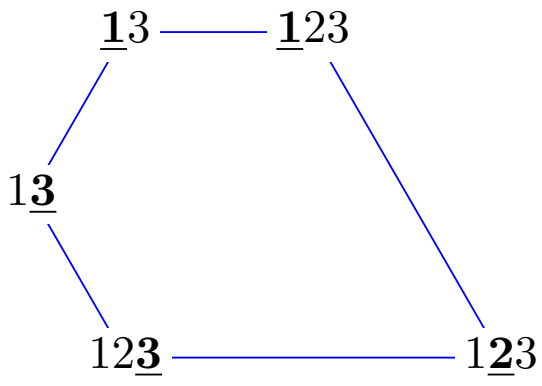
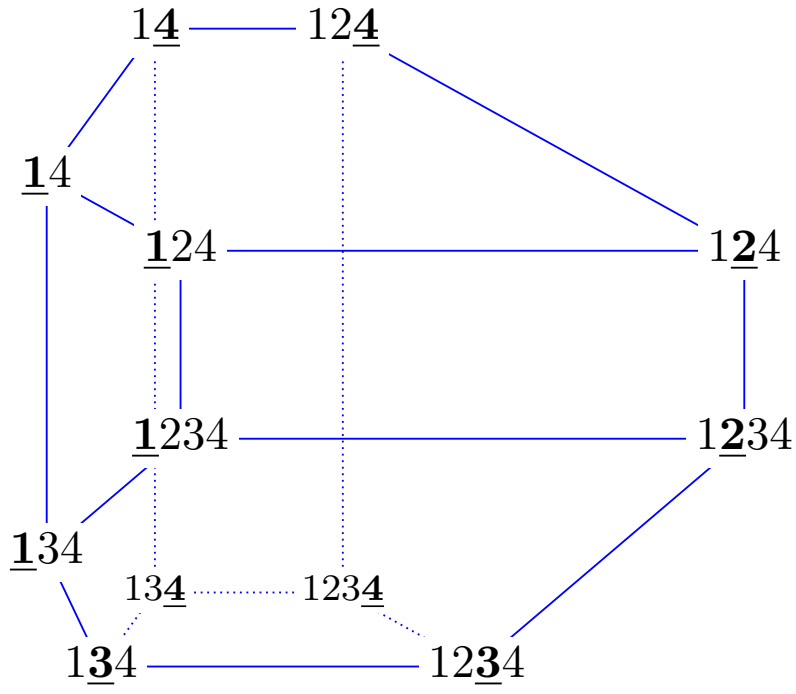
Hypergraphic polytopes

Germain Poullot with
J. Bastidas, F. Gélinas, V. Pilaud, A. Sack, E. Tzanaki





27 October 2026, Osnabrück arXiv:2606.18376

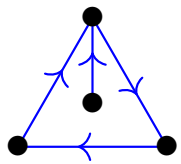
1. Acyclic orientations of hypergraphs
   - Graphs
   - Hypergraphic polytopes
   - Acyclic orientations of hypergraphs and flips

2. Graph properties of hypergraphic polytopes
   - Simple hypergraphic polytopes?
   - Distances and diameters

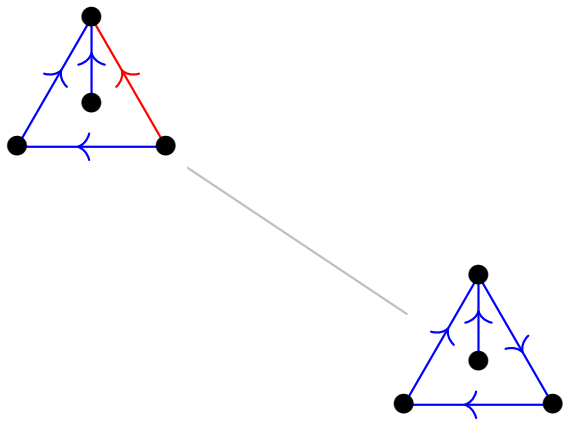
3. Combinatorics of deformed associahedra
   - Deformed associahedra
   - Tamari interval posets and weeping willows

Acyclic orientations of hypergraphs

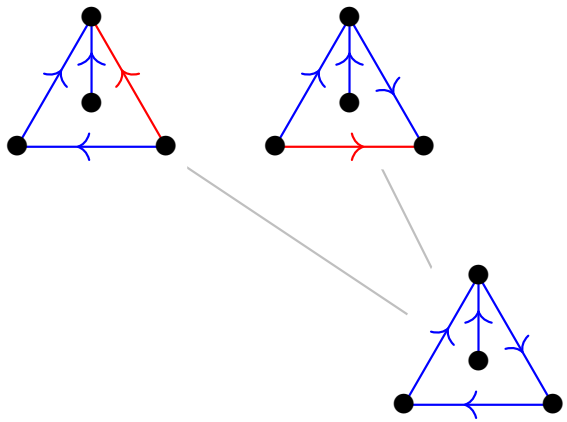
Acyclic orientations of graphs



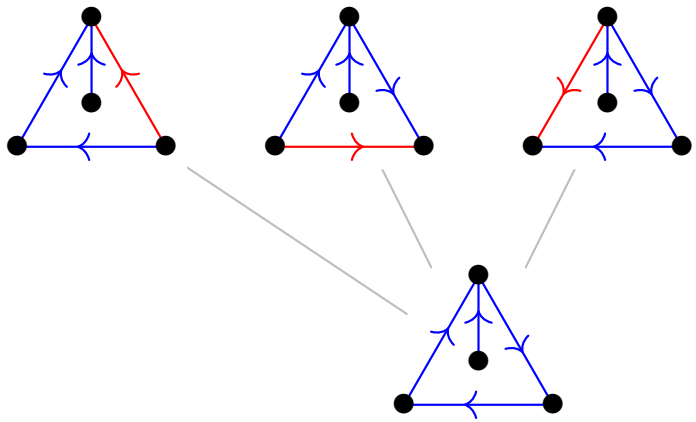
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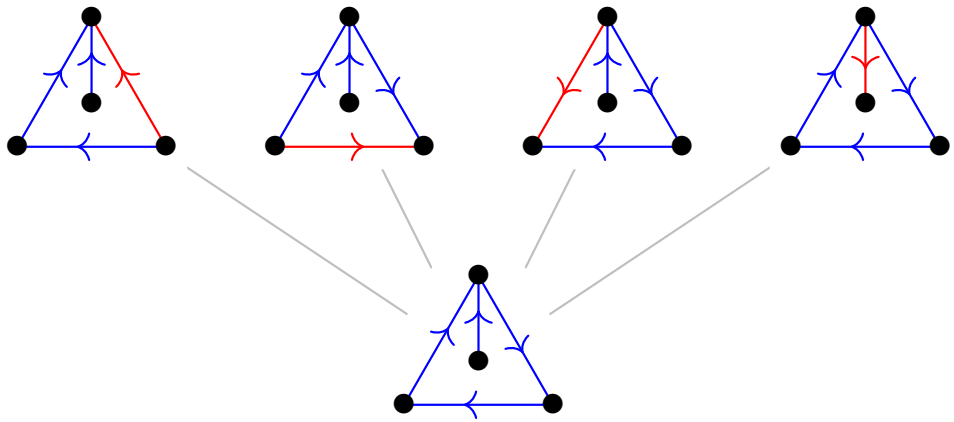
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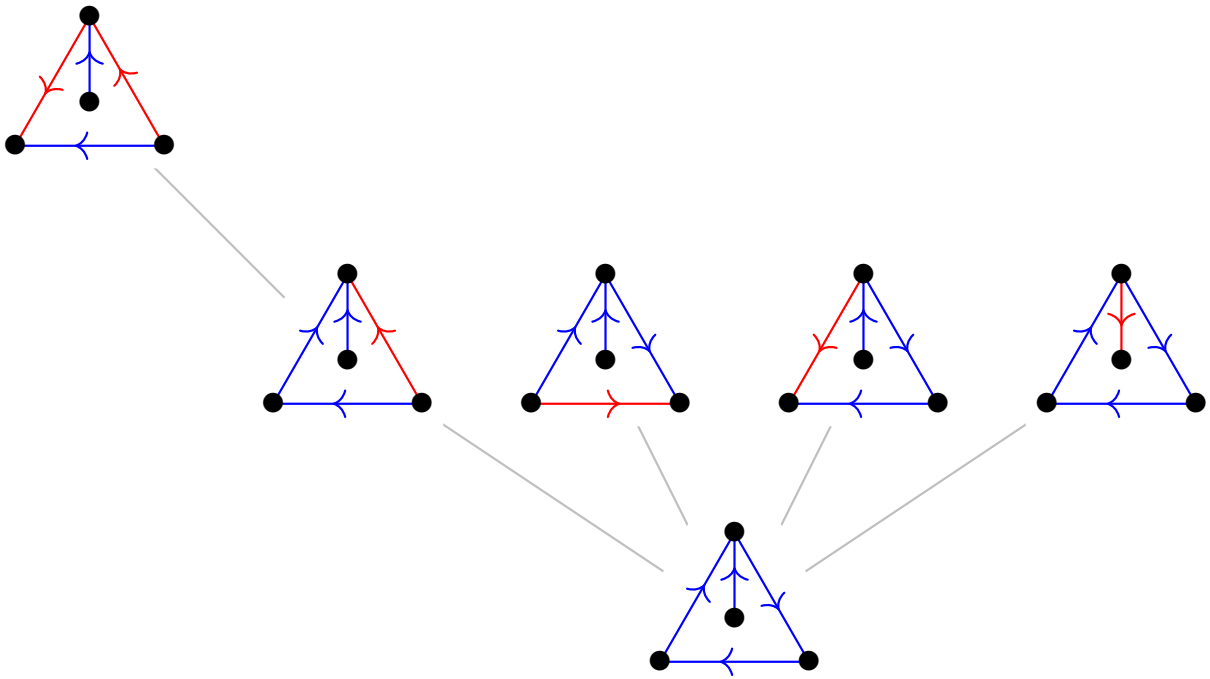
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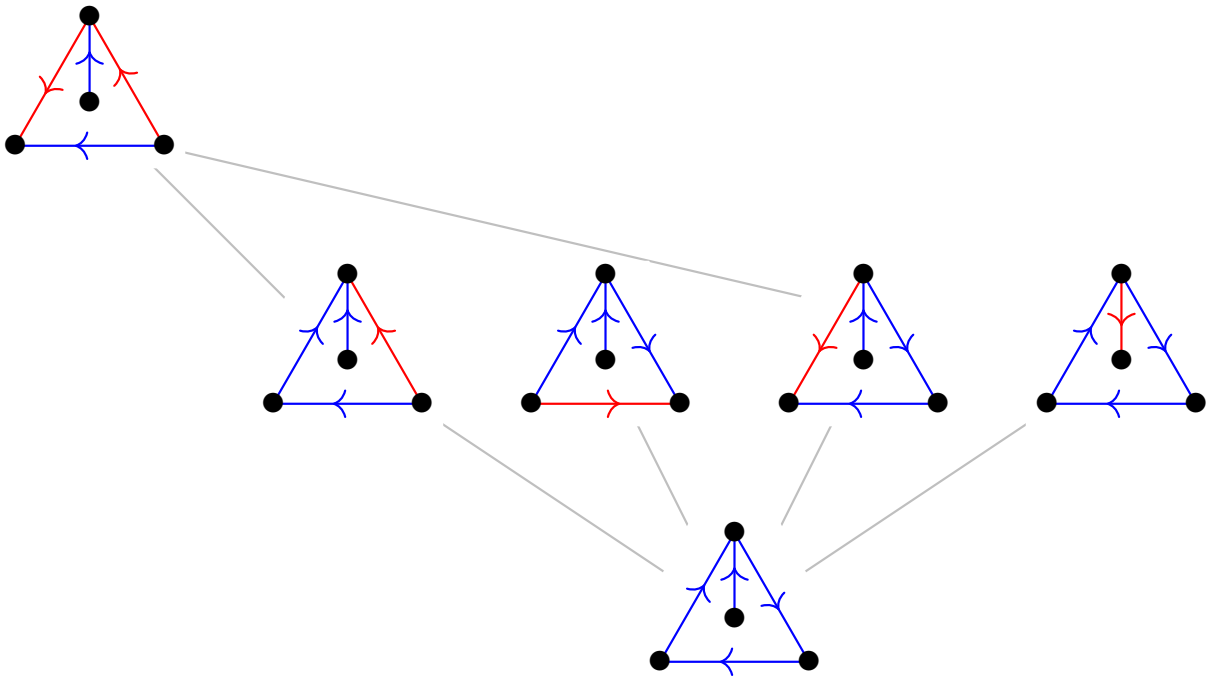
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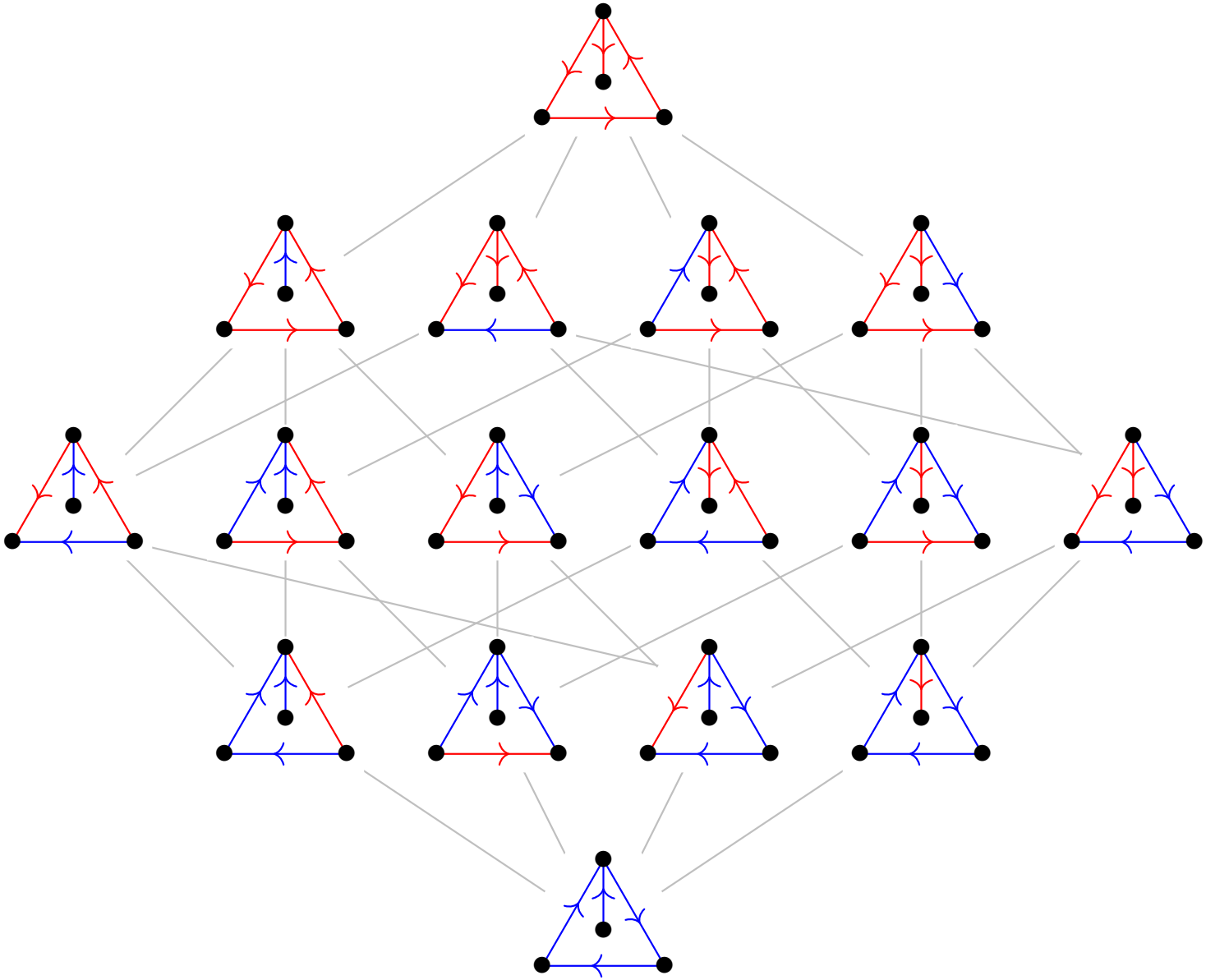
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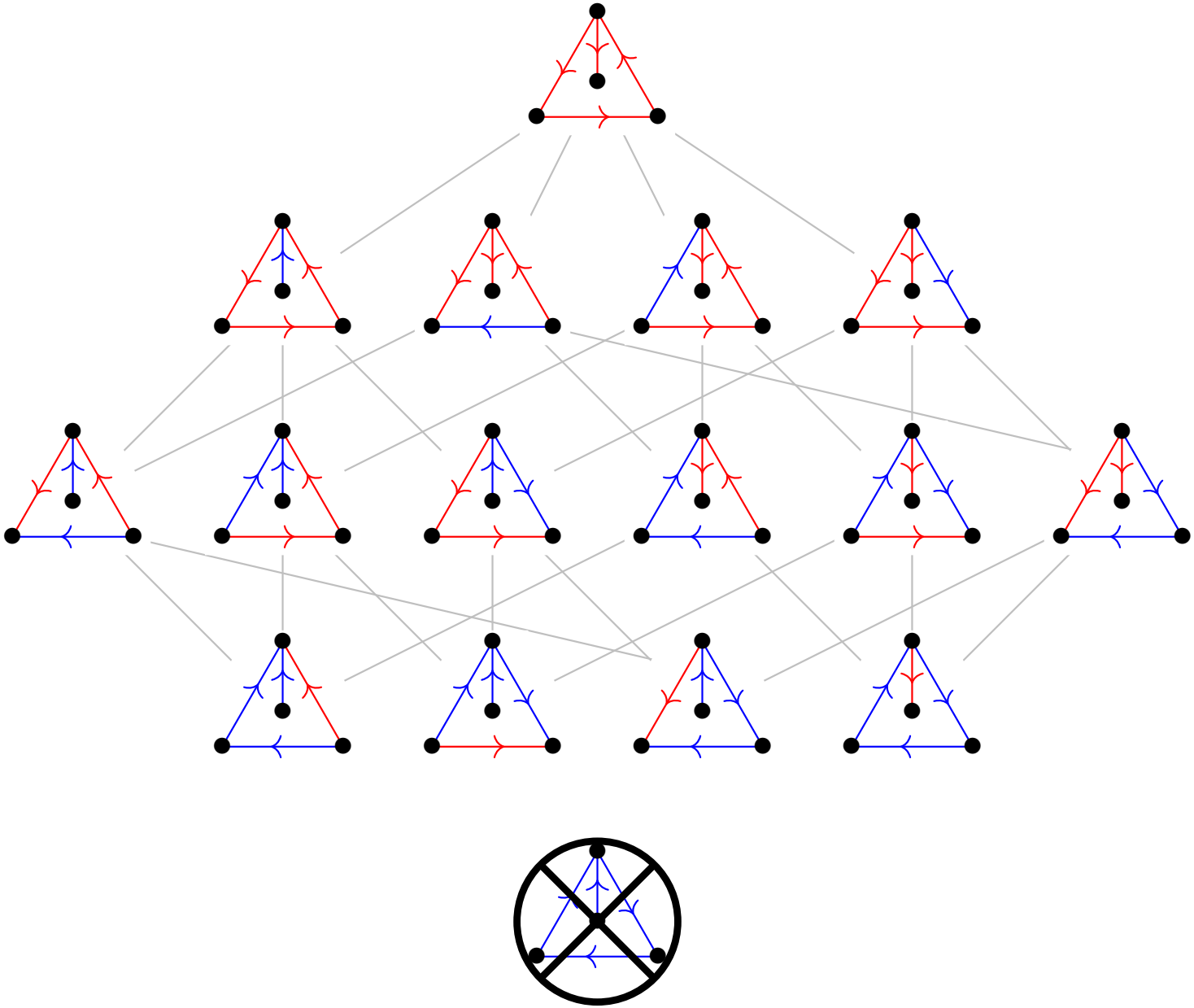
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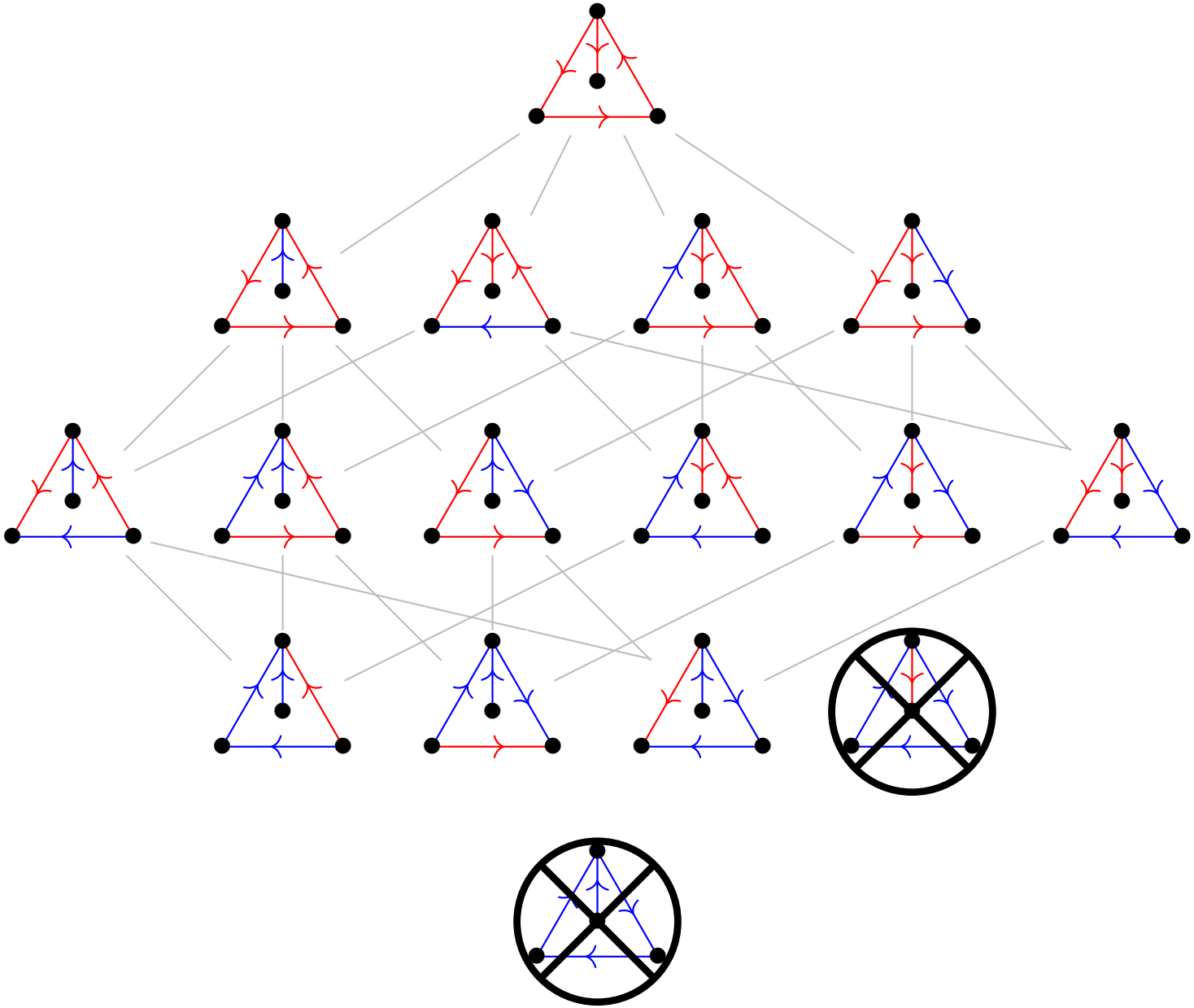
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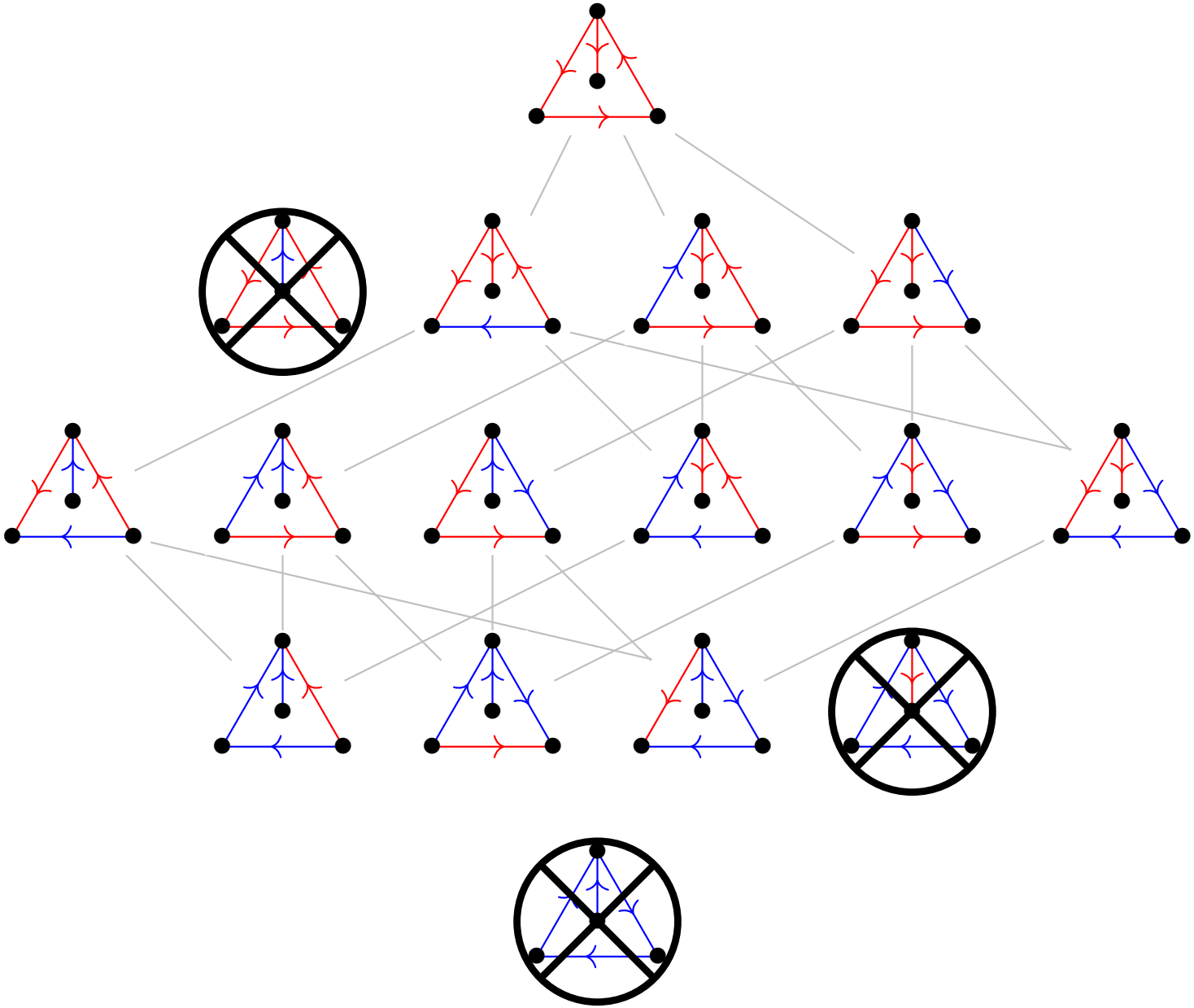
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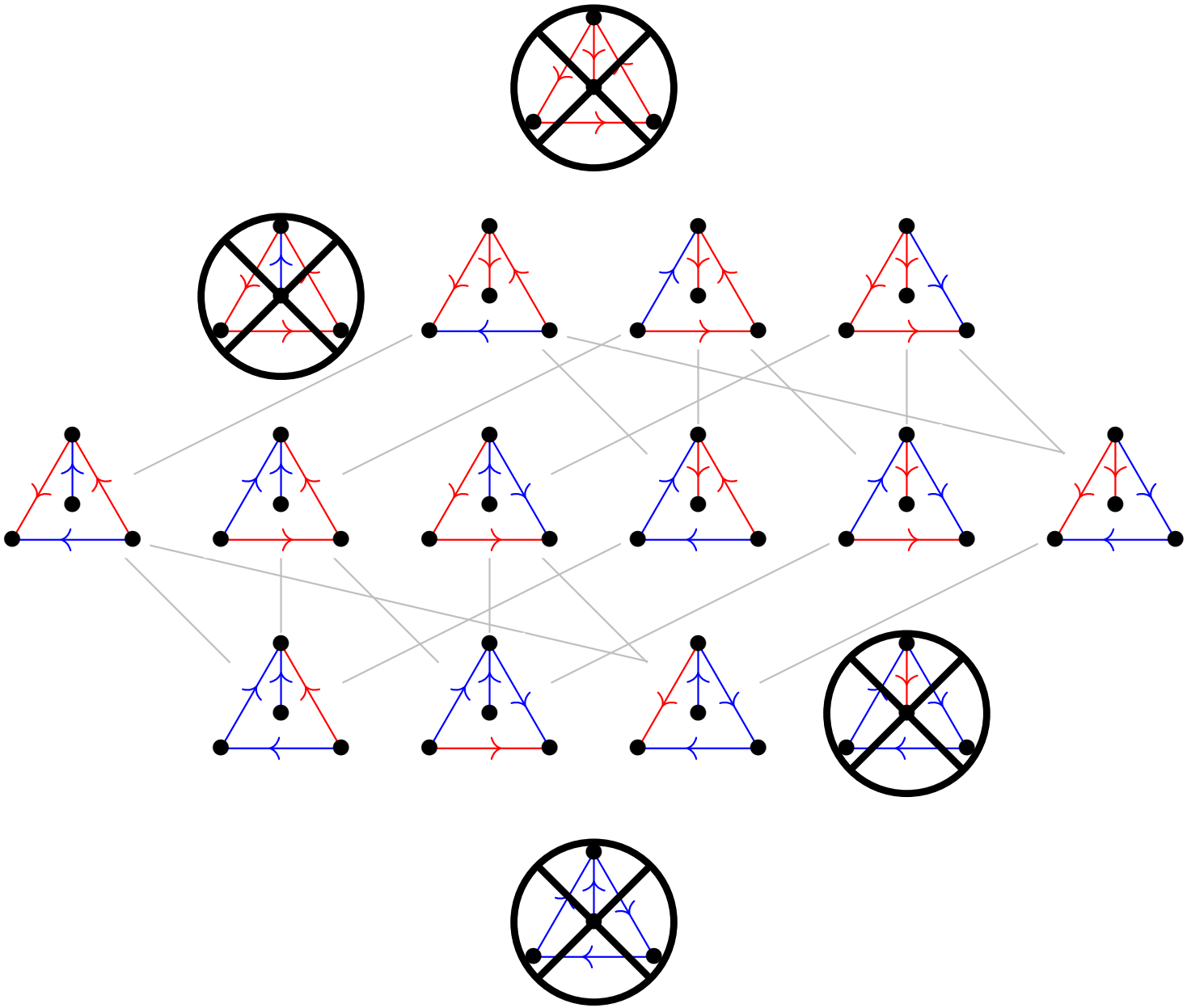
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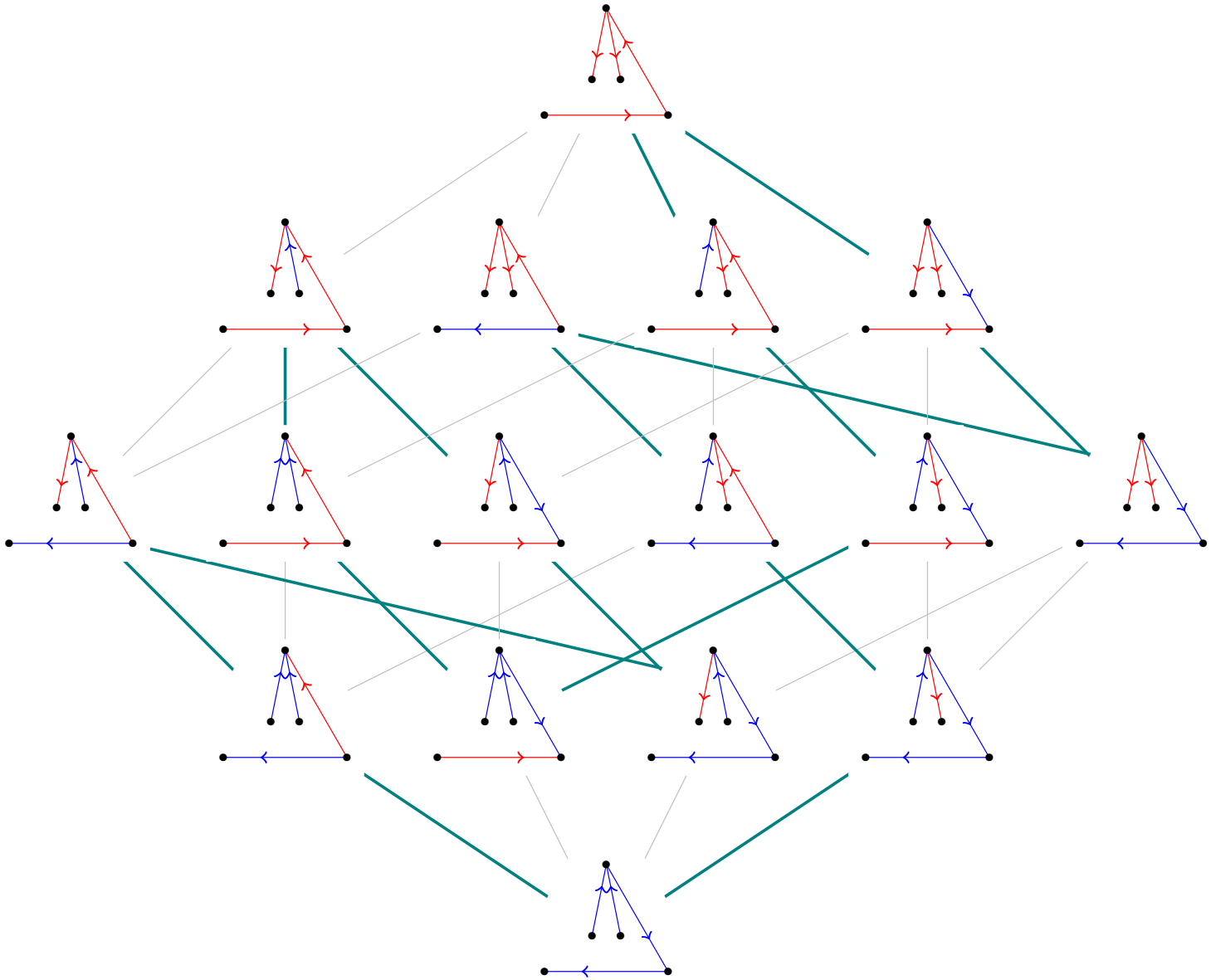
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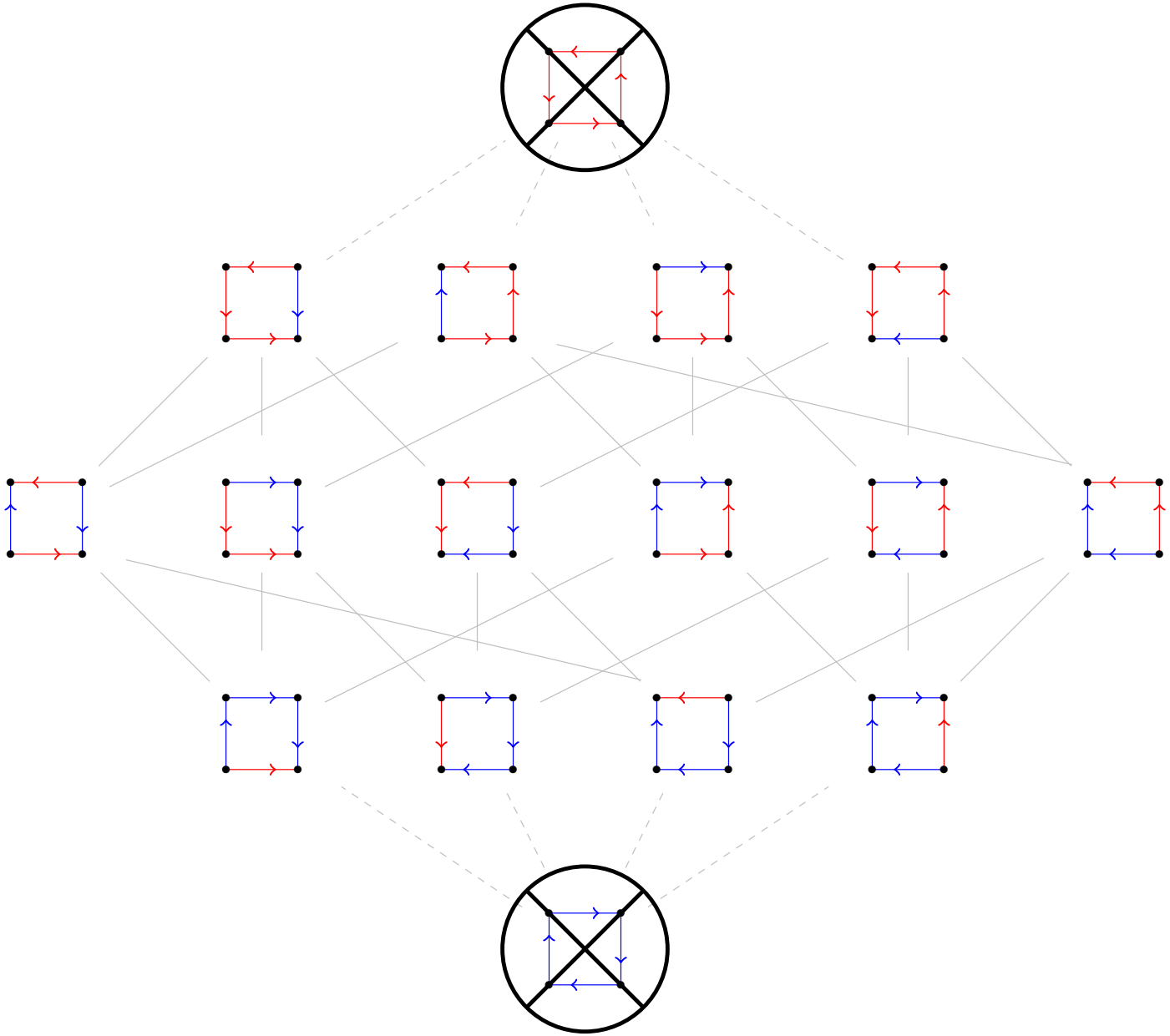
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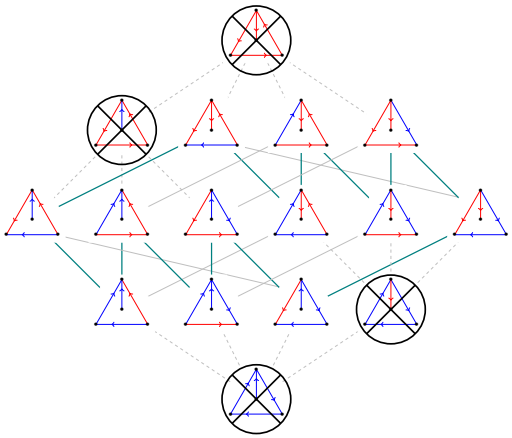
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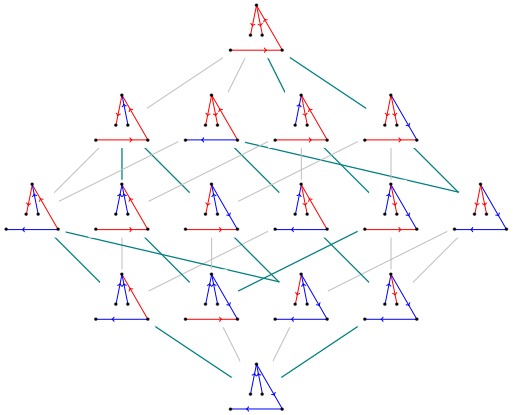


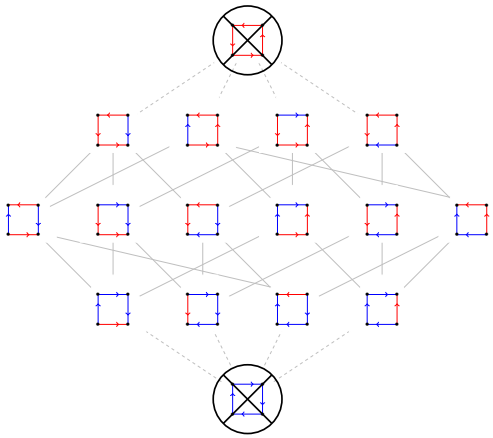
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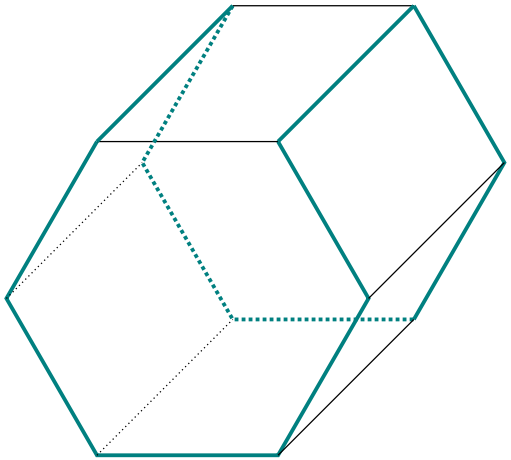


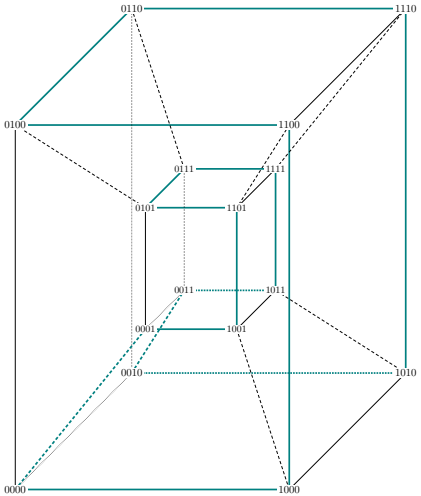
Graphical zonotopes

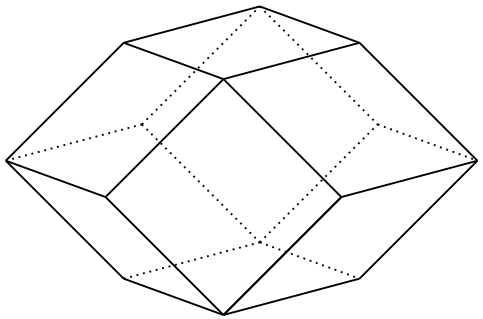












Graphical zonotopes

Minkowski sum:

$$\mathrm{P} + \mathrm{Q} = \{\boldsymbol{p} + \boldsymbol{q} \ ; \ \boldsymbol{p} \in \mathrm{P}, \ \boldsymbol{q} \in \mathrm{Q}\}$$

Definition

$G = (V, E)$ a graph.

$$Z_G = \sum_{ij \in E} [\boldsymbol{e}_i, \boldsymbol{e}_j]$$

Hypergraphic polytopes

Hypergraph: $\mathbb{H} = (V, E)$ with $V = [n]$ and $E \subseteq 2^{[n]}$

Examples:

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- interval hypergraph: for $X \in E$, we have $X = [i, j]$ for some $i \leq j$
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  - nested or disjoint: either $X \cap Y = \varnothing$ or $X \subseteq Y$ or $X \supseteq Y$

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Many more!

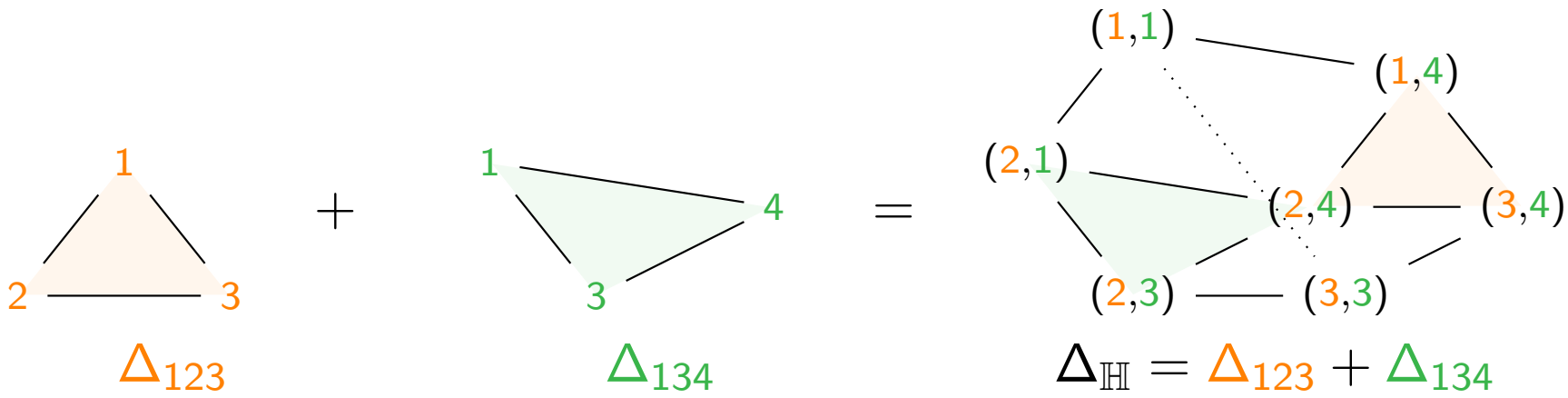
Hypergraphic polytopes

Definition (Hypergraphic polytope)

$$\Delta_{\mathbb{H}} = \sum_{X \in \mathbb{H}} \Delta_X \qquad \text{where } \Delta_X = \text{conv}\{\boldsymbol{e}_i \ ; \ i \in X\}$$

Example:

$$\mathbb{H} = \{123, 134\}$$



Δ_{123} Δ_{134} $\Delta_{\mathbb{H}} = \Delta_{123} + \Delta_{134}$

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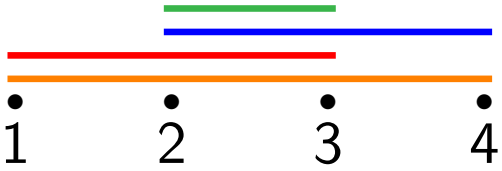
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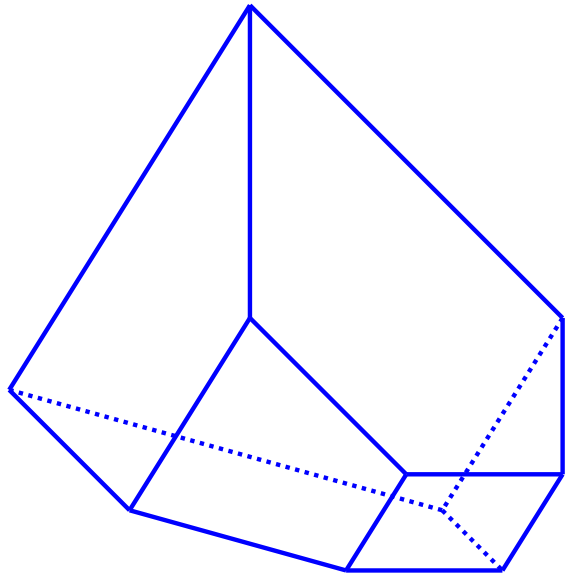
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- graphical zonotope of its 1-skeleton

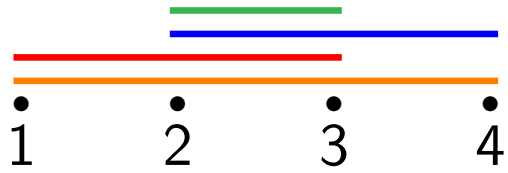
Many more! e.g. parking-function polytope, master polytope, ...

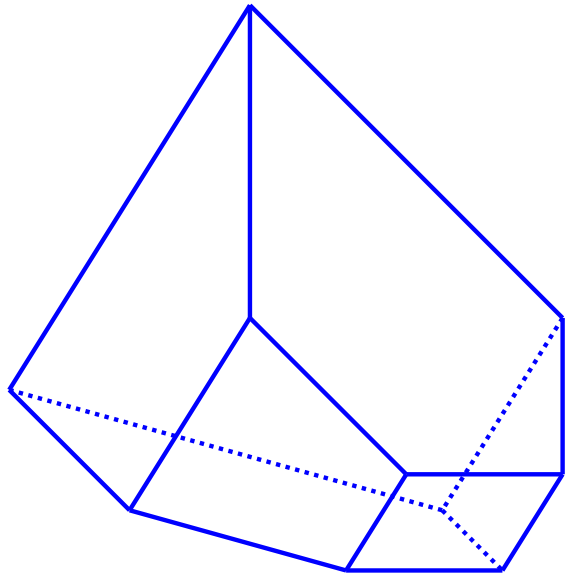
Acyclic orientation of hypergraphs





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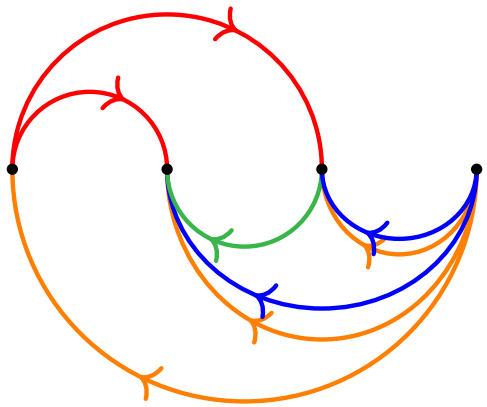




Orientation $O : \mathbb{H} \to [n]$ with $O(X) \in X$

Graph of sources: directed graph $O(X) \to y$ for $y \in X \setminus \{O(X)\}$

Acyclic orientation: graph of source is acyclic



Acyclic orientation of hypergraphs

Theorem

Vertices of $\Delta_{\mathbb{H}} \longleftrightarrow$ *Acyclic orientations of* $\mathbb{H}$

Edges of $\Delta_{\mathbb{H}} \longleftrightarrow$ *???*

Acyclic orientation of hypergraphs

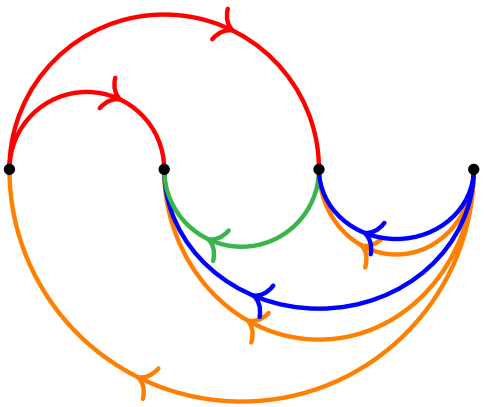
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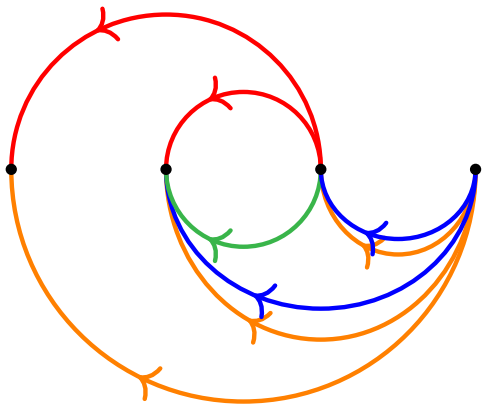
Edges of $\Delta_{\mathbb{H}} \longleftrightarrow$ *???*

A acyclic orientations, $a, b \in [n]$. *flip* of A at (a, b):

$$B(X) = \begin{cases} b & \text{if} A(X) = a \text{ and } b \in X \\ A(X) & \text{otherwise} \end{cases}$$



A



flip of A at $(1, 3)$

Acyclic orientation of hypergraphs

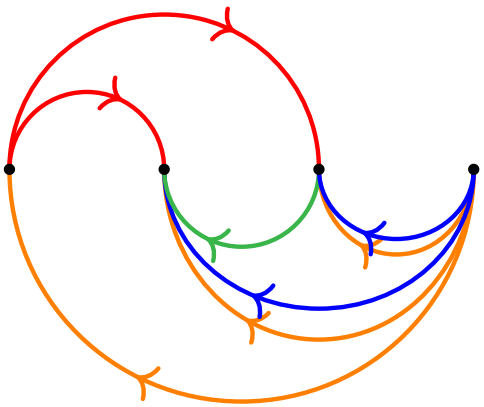
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Vertices of $\Delta_{\mathbb{H}} \longleftrightarrow$ *Acyclic orientations of* $\mathbb{H}$

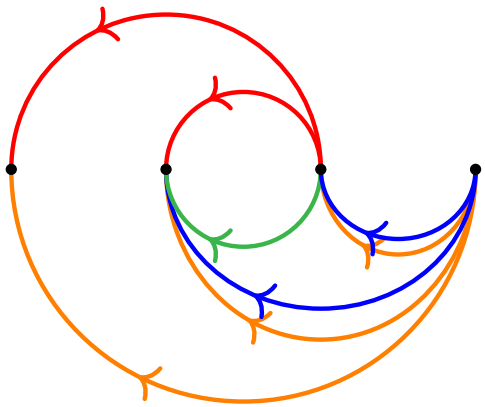
Edges of $\Delta_{\mathbb{H}} \longleftrightarrow$ *Flips between acyclic orientations*

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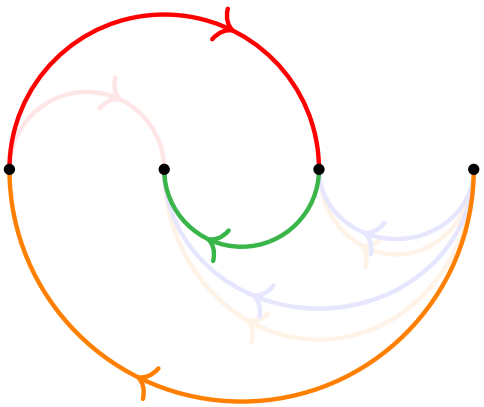
Theorem

Vertices of $\Delta_{\mathbb{H}} \longleftrightarrow$ *Acyclic orientations of* $\mathbb{H}$

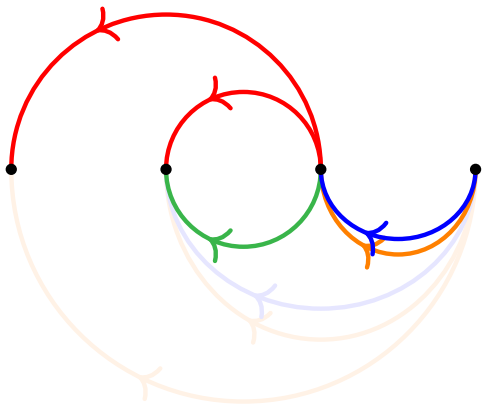
Edges of $\Delta_{\mathbb{H}} \longleftrightarrow$ *Flips between acyclic orientations*

A acyclic orientations, $a, b \in [n]$. *flip* of A at (a, b):

$$B(X) = \begin{cases} b & \text{if} A(X) = a \text{ and } b \in X \\ A(X) & \text{otherwise} \end{cases}$$



A



flip of A at $(1, 3)$

transitive reduction of A: arcs $i \to j$ of the graph of source such that there is no path of length ≥ 2 from i to j

Graph properties of hypergraphic polytopes

Simple hypergraphic polytopes

Simple polytope: at each vertex of P, $|\text{adjacent edges}| = \dim \mathrm{P}$

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Theorem (BGPP**ST '26)**

$\mathbb{I}$: *interval hypergraph*

Explicit criterion for $\Delta_{\mathbb{I}}$ *to be simple, computable in time* $O(|\mathbb{I}|^4)$

Simple hypergraphic polytopes

For hypergraphic polytope $\Delta_{\mathbb{H}}$:
"For all acyclic orientation of $\mathbb{H}$, is $|\text{transitive reduction}| = n - 1$?"

Open question:

Can this problem be solved in polynomial time for all hypergaphs?

You can solve "**P** = **NP**" if you prefer.

About distance

A, B acyclic orientations
$A \triangle B = \{X \in \mathbb{H} \ ; \ A(X) \neq B(X)\}$
$A(A \triangle B) = \{A(X) \ ; \ X \in A \triangle B\}$

Theorem (Gélinas–P. '26+)

$d(A, B)$ flip-distance between A and B

$$\max\{|A(A \triangle B)|, |B(A \triangle B)|\} \ \leq \ d(A, B) \ \leq \ |A \triangle B|$$

Proof idea:

About distance

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Corollary (Gélinas–P. '26)

$$\mathit{diameter}(\Delta_{\mathbb{H}}) \leq \min\{|\mathbb{H}|, \tbinom{n}{2}\}$$

Hyper-permutahedra

$\binom{[n]}{k}$ all subsets of size k

k^{th} *hyper-permutahedron*: $\Delta_{\binom{[n]}{k}}$

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The distance on hyper-permutahedra can be computed in polynomial time, and

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Board explanations with a magic hat

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If $\mathbb{H} = \binom{[n]}{k} \cup \mathbb{H}'$ where $|X| \geq k+1$ for $X \in \mathbb{H}'$, then $\mathit{diameter}(\Delta_{\mathbb{H}}) = \binom{n-k+2}{2}$.

Freehedon

$\mathbb{IF}_n$ = all initial $[i]$ and final $[n] \smallsetminus [i]$ intervals

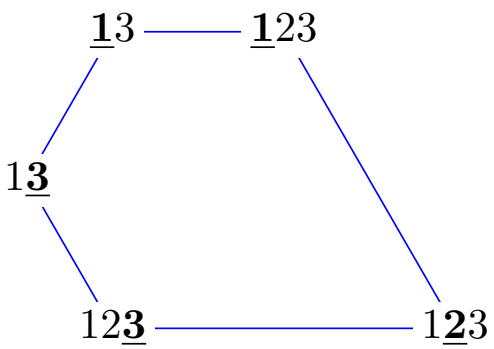
freehedron $\mathsf{Free}_n = \Delta_{\mathbb{IF}_n}$

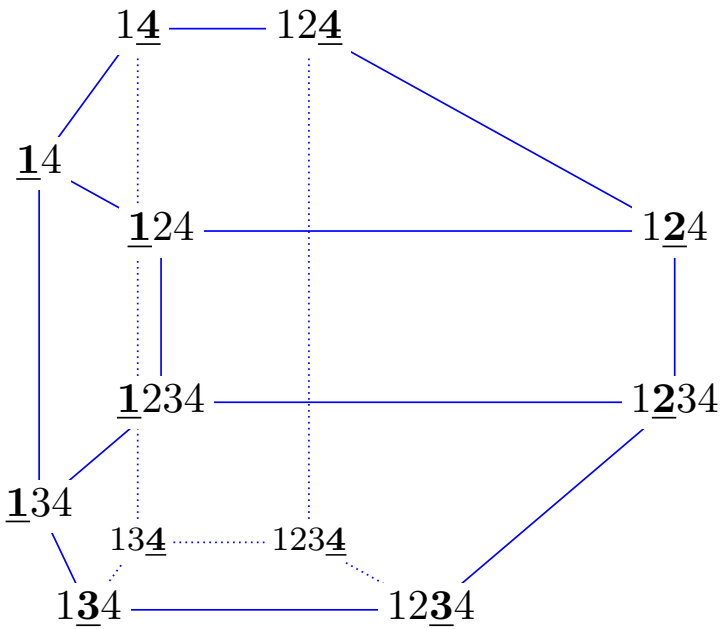
Theorem (Gélinas–P. '26)

The distance on hyper-permutahedra can be computed in polynomial time, and

$$\mathit{diameter}(\mathsf{Free}_n) = n - 1$$

Vertices of $\mathsf{Free}_n \longleftrightarrow$ "pointed sequence"





Many examples

	Nom	Diameter	Reference
$\mathbb{G}$	Graphical zonotope	$\lvert\mathbb{G}\rvert$	Folklore
n	Pitman–Stanley pol.	$n-1$	easy from SP '02

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1) For which (families of) hypergraphs the diam. is linear/quadratic.

2) Can the distance on the associahedron be computed in polynomial time?

Combinatorics of deformed associahedra

Deformed associahedra

Usual permutahedron:

$$\mathrm{Permuto}_n = \sum_{1 \leq i < j \leq n} [\boldsymbol{e}_i, \boldsymbol{e}_j] = \sum_{X \subseteq [n]} \Delta_X$$

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$$\text{Asso}_n = \sum_{1 \leq i < j \leq n} \Delta_{[i,j]}$$

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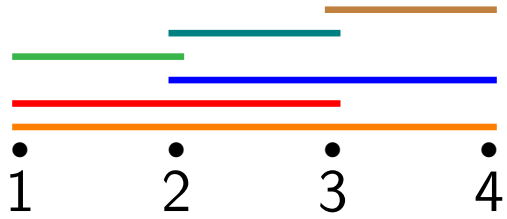
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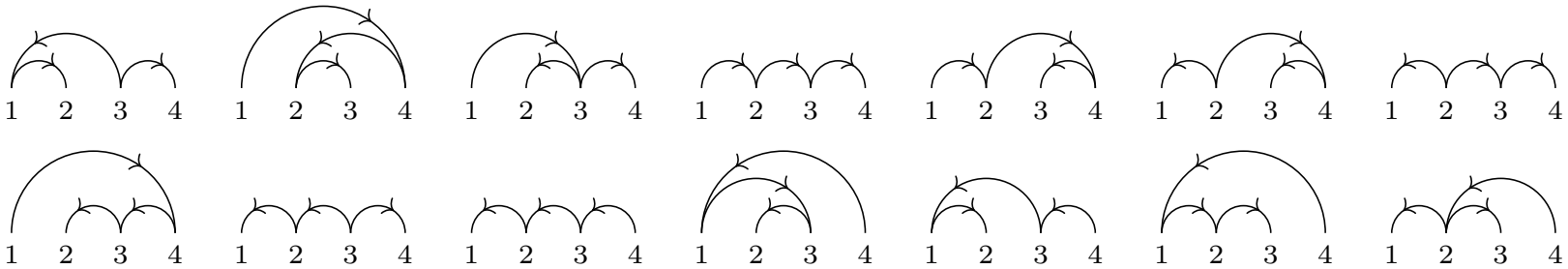
Definition

$\mathbb{I}_n = \{[i,j] \ ; \ 1 \leq i < j \leq n\}$
Interval hypergraph: $\mathbb{J} \subseteq \mathbb{I}_n$
Deformed associahedron or *interval hypergraphic polytope*: $\Delta_{\mathbb{J}}$

Combinatorics of the associahedron

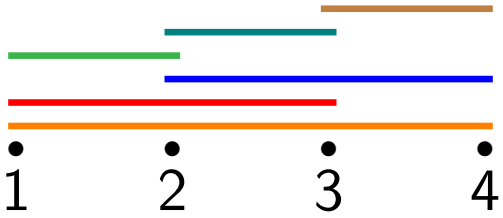
Vertices $\mathrm{Asso}_n \longleftrightarrow$ rooted binary trees

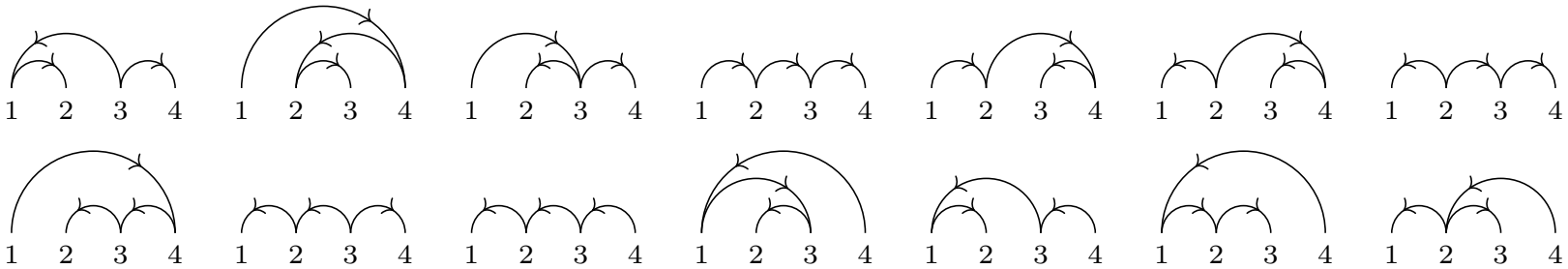




Combinatorics of the associahedron

Vertices $\mathrm{Asso}_n \longleftrightarrow$ rooted binary trees





Vertices deformed asso. $\longleftrightarrow$ some binary-tree-like combinatorics

Vertex digraphs

Remember:

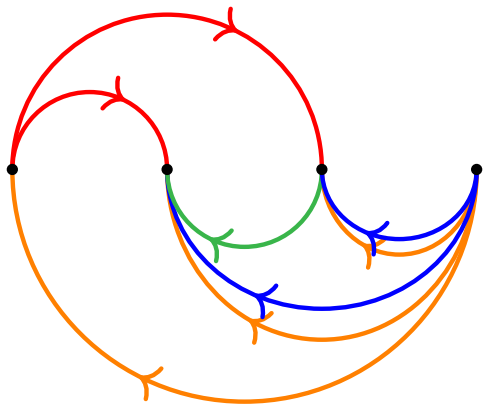
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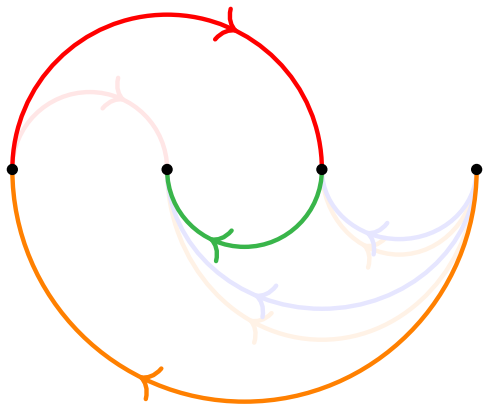


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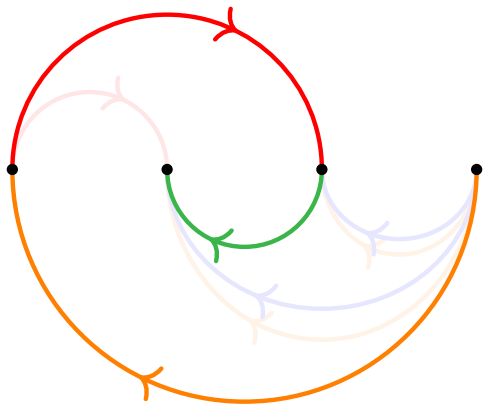


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Equivalently *vertex digraph*:

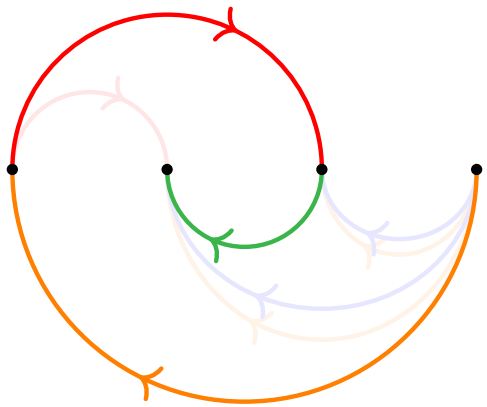
Look at a vertex $\boldsymbol{v} \in \Delta_{\mathbb{H}}$

Adjacent edges at $\boldsymbol{v}$: each one is in a direction $\boldsymbol{e}_i - \boldsymbol{e}_j$ for some i, j

Draw the arcs $i \to j$

Vertex digraphs

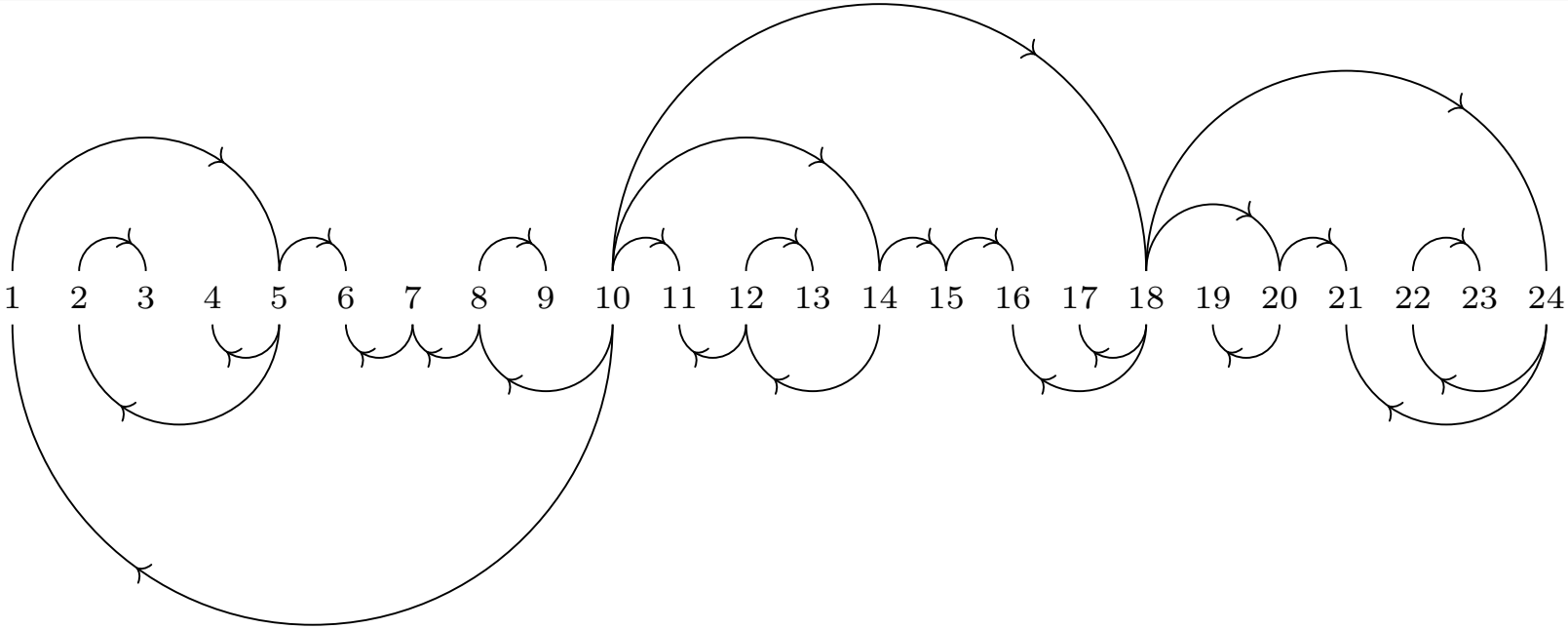
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vertex poset $\lhd =$ transitive closure of vertex digraph

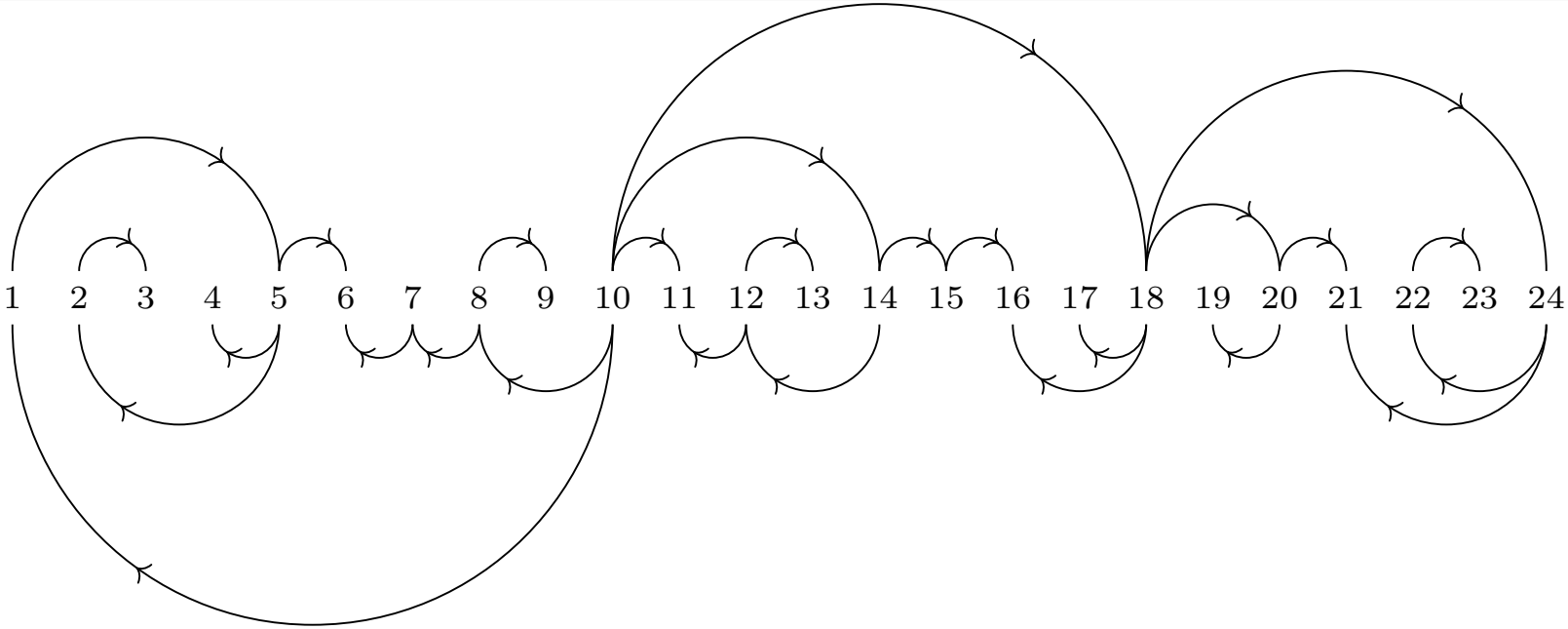
Tamari interval posets



Tamari interval posets: poset on $[n]$ such that

For $a < b < c$, if $a \lhd c$, then $a \lhd b$; and if $c \lhd a$, then $c \lhd a$

Tamari interval posets



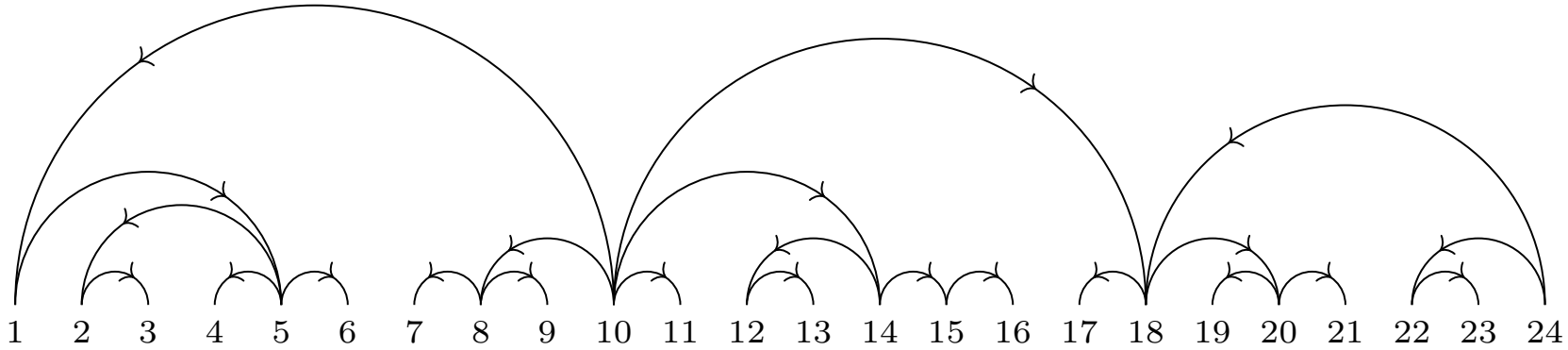
Tamari interval posets: poset on $[n]$ such that
For $a < b < c$, if $a \triangleleft c$, then $a \triangleleft b$; and if $c \triangleleft a$, then $c \triangleleft a$

Theorem (BGPP**ST '26)**

$\{$all vertices of all deformed associahedra$\} \leftrightarrow \{$Tamari int. posets$\}$
all vertex posets of $\Delta_{\mathbb{H}}$ are Tamari int. posets $\Rightarrow$ $\Delta_{\mathbb{H}}$ is a deformed associahedron

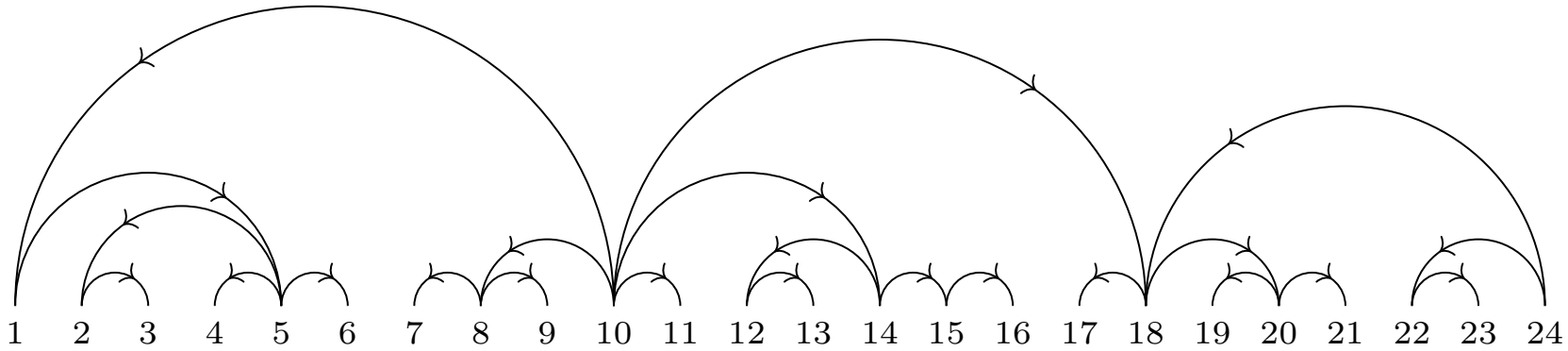
Actually, applies to deformed permutahedra

Weeping willows



Weeping willow: tree on $[n]$ such that
For $a < b < c$, if $a \to c$, then $\exists$ path $a \rightsquigarrow b$; and if $c \to a$, then $\exists$ path $c \rightsquigarrow b$

Weeping willows



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Theorem (BGPPST **'26)**

*$\{$all vertices of all **simple** deformed asso.$\}$ $\leftrightarrow$ $\{$weeping willows$\}$*
all vertex posets of $\Delta_{\mathbb{H}}$ are weeping willows $\Rightarrow$ $\Delta_{\mathbb{H}}$ is a simple deformed associahedron

Actually, applies to deformed permutahedra

Counting!

\# Tamari interval posets [Chapoton '07]

$$\frac{2}{(3n+1)(3n+2)}\binom{4n+1}{n+1}$$

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Rooted weeping willows [BGP**P**ST '26]

$$\frac{1}{n+1}\binom{3n+1}{n}$$

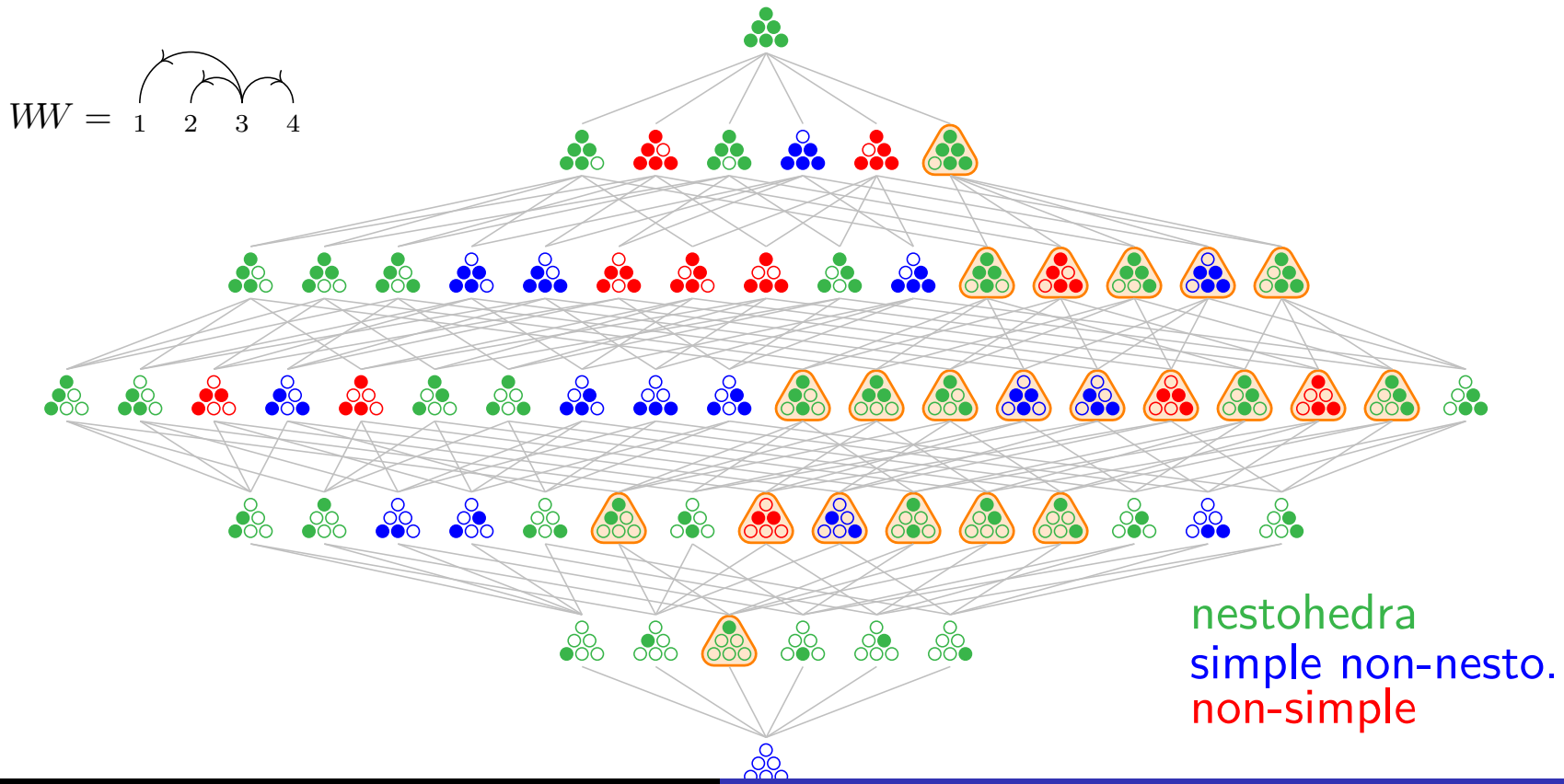
Many more variants...

Finer description

Given a Tamari int. poset $\triangleleft$ (resp. a weeping willow), which are the hypergraphs $\mathbb{H}$ such that $\triangleleft \in \{\text{vertices of } \mathbb{H}\}$?

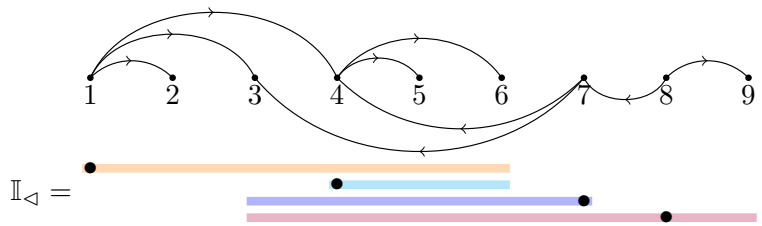
Theorem (BGPPST **'26)**

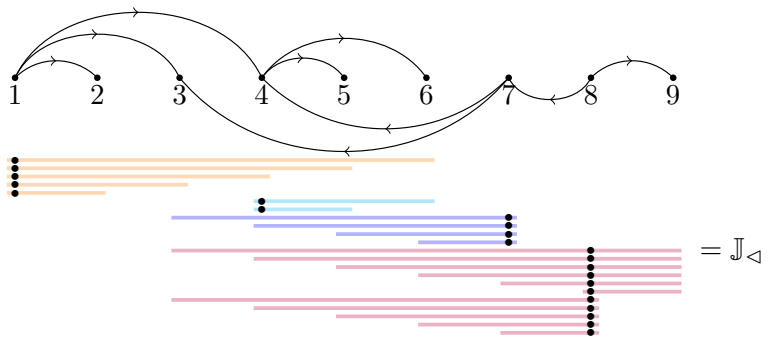
$\mathcal{I}_\triangleleft$ is order convex in $\mathcal{I}$.
$\mathcal{S}_\triangleleft$ is order convex in $\mathcal{S}$, not in $\mathcal{I}$.



Finer description

$\mathbb{I}_\lhd = \{\text{maximal descendant sets}\}$
$\mathbb{J}_\lhd = \{\text{all descendant sets}\}$





Theorem (BGPPST **'26)**

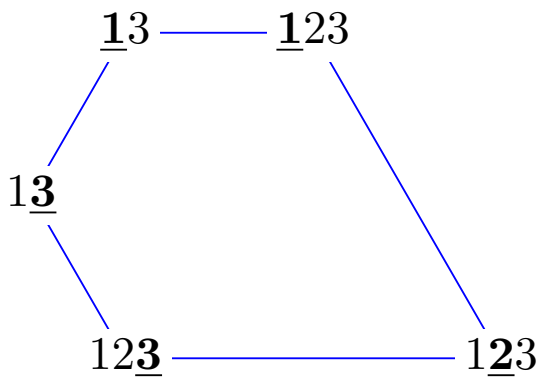
$\mathbb{I}_\lhd$ is a minimum of $\mathcal{I}_\lhd$ + condition for unicity
*$\mathbb{J}_\lhd$ is the **unique** maximum of $\mathcal{I}_\lhd$*

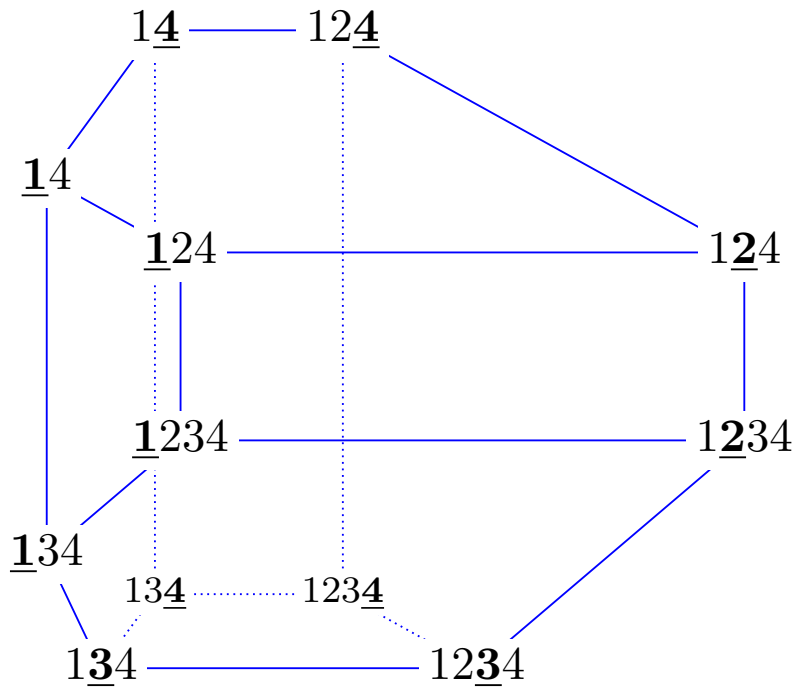
Going further beyond

Stuffs I did not talk about:

- Deformations of hypergraphic polytopes
- Hypergraphic posets (Hasse diagram, lattices, ...)
- Tropical oriented matroids
- Ehrhart theory, regular unimodular triangulations, ...
- Plumbing bijections
- Hypergraph saturation
- Faces of interval hypergraphic polytopes
- ...

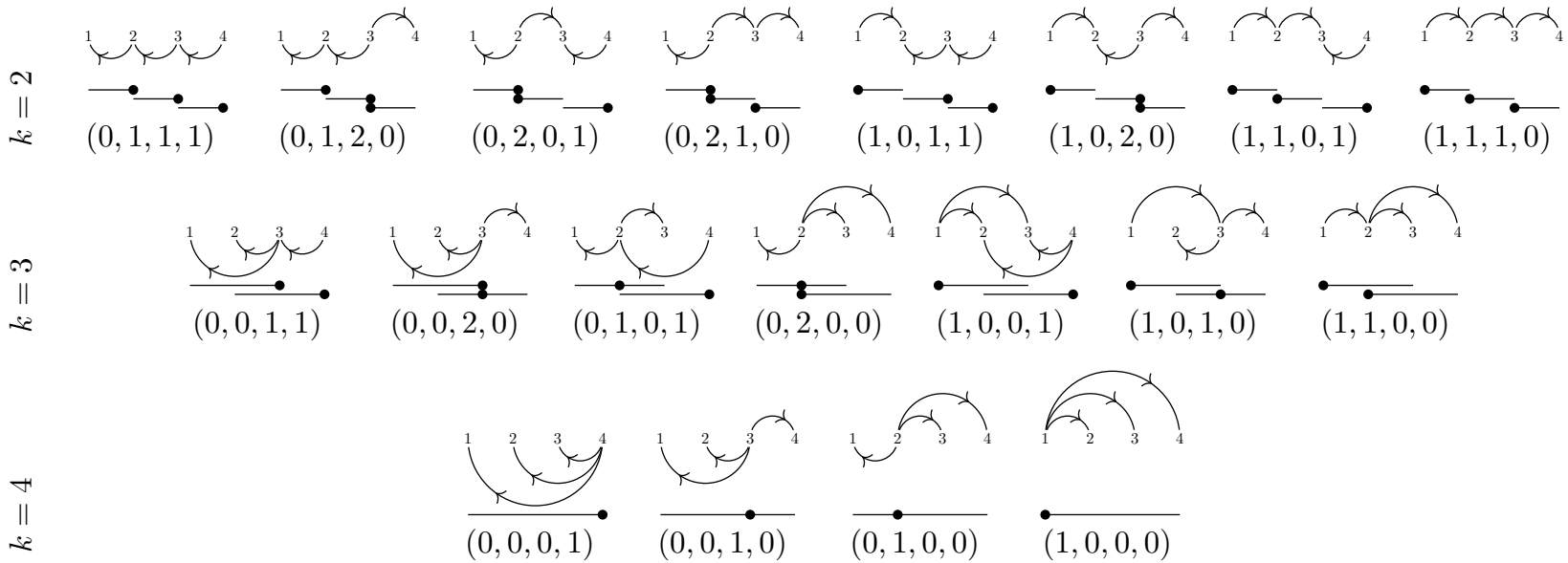
Thank you!


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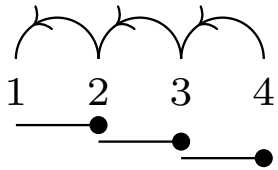
Bonus: Tamari interval posets

Complete uniform interval hypergraph: posets and vertices:

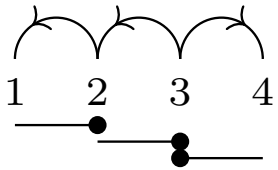


Bonus: Weeping willows

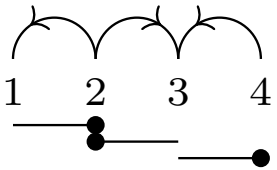
Path-graph zonotope (i.e. cube) trees and vertices:



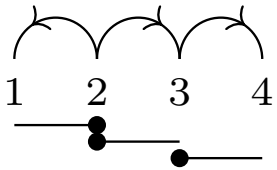
(0,1,1,1)



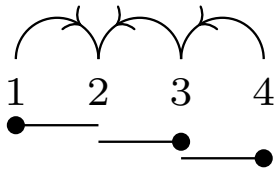
(0,1,2,0)



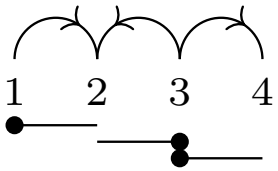
(0,2,0,1)



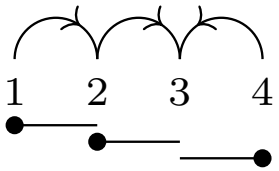
(0,2,1,0)



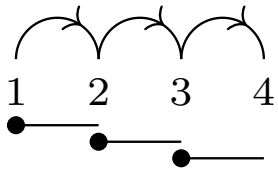
(1,0,1,1)



(1,0,2,0)



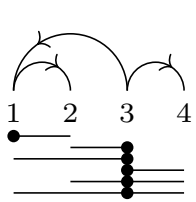
(1,1,0,1)



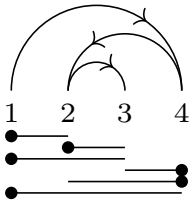
(1,1,1,0)

Bonus: Weeping willows

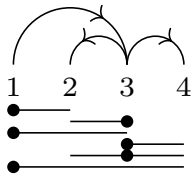
Associahedron trees and vertices:



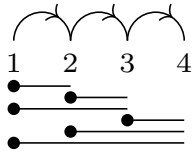
(1,0,5,0)



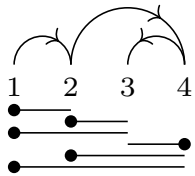
(3,1,0,2)



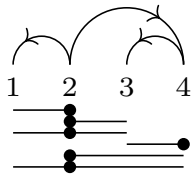
(3,0,3,0)



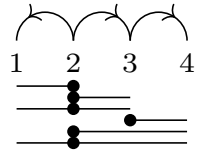
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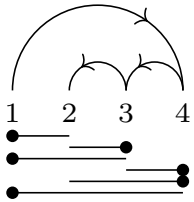
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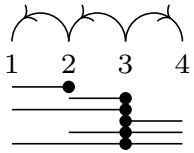
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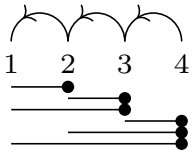
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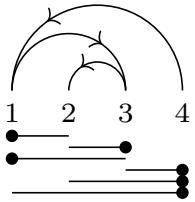
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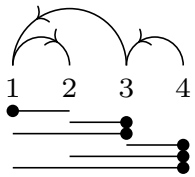
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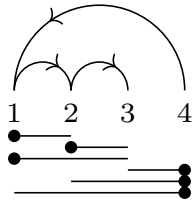
(0,1,2,3)



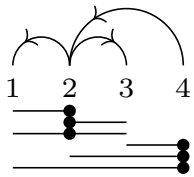
(2,0,1,3)



(1,0,2,3)



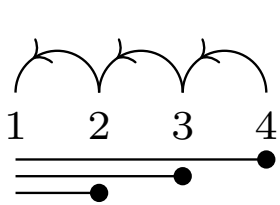
(2,1,0,3)



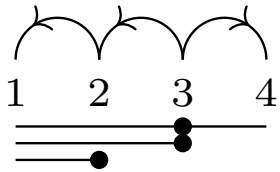
(0,3,0,3)

Bonus: Weeping willows

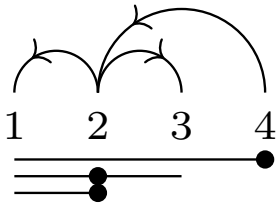
Pitman–Stanley trees and vertices:



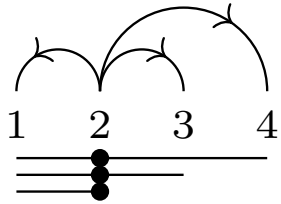
(0,1,1,1)



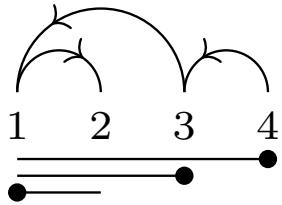
(0,1,2,0)



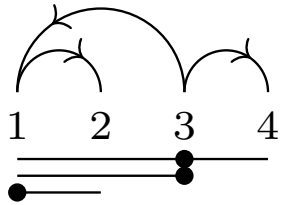
(0,2,0,1)



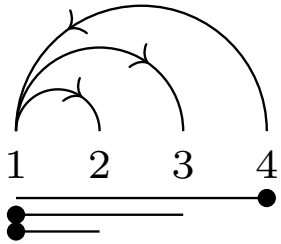
(0,3,0,0)



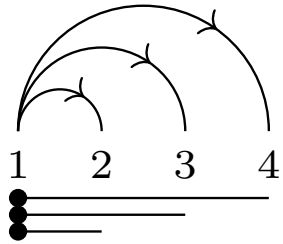
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(1,0,2,0)



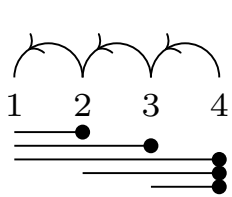
(2,0,0,1)



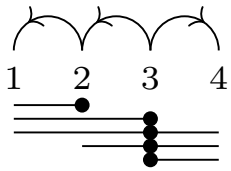
(3,0,0,0)

Bonus: Weeping willows

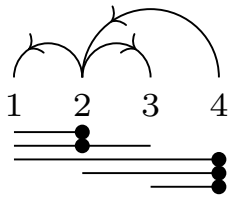
Freehedron trees and vertices:



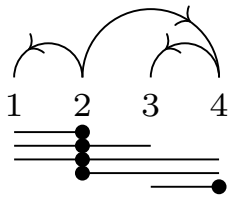
(0,1,1,3)



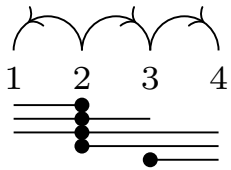
(0,1,4,0)



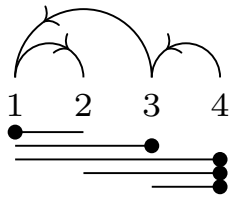
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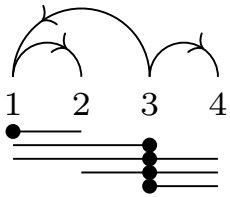
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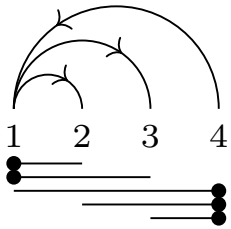
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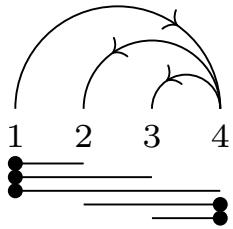
(1,0,1,3)



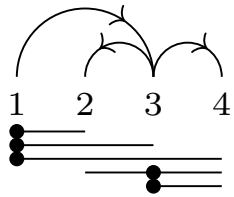
(1,0,4,0)



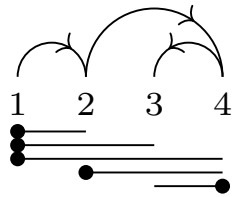
(2,0,0,3)



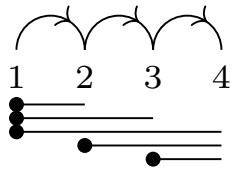
(3,0,0,2)



(3,0,2,0)



(3,1,0,1)



(3,1,1,0)